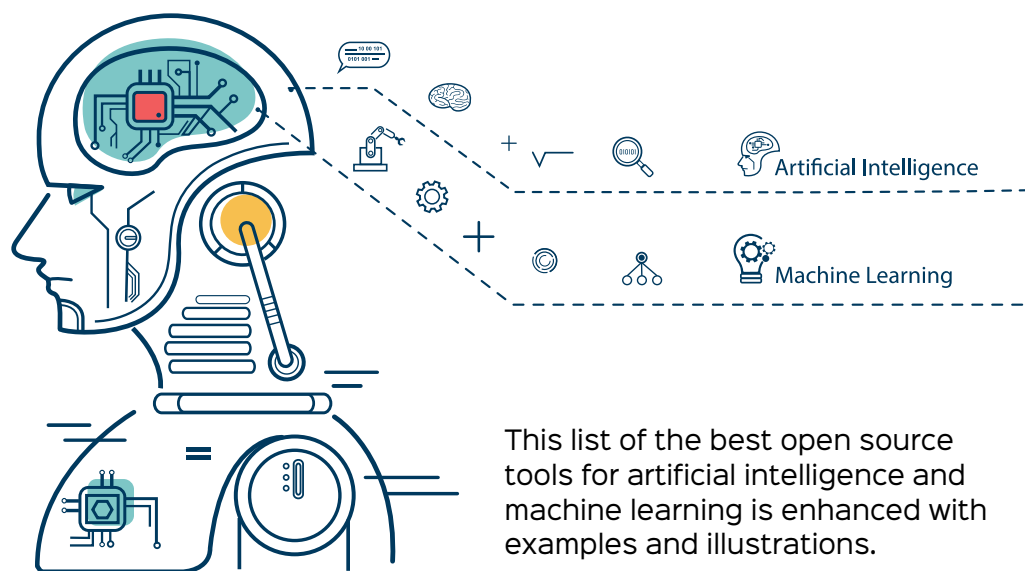


Top 10 Open Source Tools for AI/ML



This list of the best open source tools for artificial intelligence and machine learning is enhanced with examples and illustrations.

Artificial intelligence (AI) and machine learning (ML) drive innovation across industries, and open source tools play a pivotal role in democratising access to cutting-edge technologies. These tools provide flexibility, community support, and rapid development capabilities. Here is a comprehensive list of top open source tools for AI and ML, enriched with examples and illustrations.

1 TensorFlow

TensorFlow, developed by Google, is one of the most widely used libraries for deep learning and machine learning. It supports a range of tasks, from image recognition to time-series forecasting.

Example: Building a simple neural network for image classification:

```
import tensorflow as tf
from tensorflow.keras import layers, models

model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_
shape=(28, 28, 1)),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
```

```
layers.Dense(64, activation='relu'),
layers.Dense(10, activation='softmax')
```

```
])
```

Key features:

- Scalable across multiple GPUs and distributed systems.
- Supports TensorFlow Lite for mobile and embedded devices.
- Extensive documentation and pre-trained models.

Use cases:

- Image classification
- Natural language processing

2 PyTorch

Developed by Facebook, PyTorch is a popular deep learning framework known for its dynamic computational graph and user-friendly interface.

Example: Training a basic neural network with PyTorch:

```
import torch
import torch.nn as nn

class SimpleNN(nn.Module):
```